

Linear Algebra

HSLU, Semester 4

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1 Vectors

1.1 Vector arithmetic

Let $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$. Then:

$$\begin{aligned}\vec{a} + \vec{b} &= \vec{b} + \vec{a}, & (\vec{a} + \vec{b}) + \vec{c} &= \vec{a} + (\vec{b} + \vec{c}), \\ \vec{a} + \vec{0} &= \vec{a}, & \vec{a} + (-\vec{a}) &= \vec{0}, \\ \lambda(\vec{a} + \vec{b}) &= \lambda\vec{a} + \lambda\vec{b}, & (\lambda + \mu)\vec{a} &= \lambda\vec{a} + \mu\vec{a}, \\ (\lambda\mu)\vec{a} &= \lambda(\mu\vec{a}), & 1\vec{a} &= \vec{a}.\end{aligned}$$

For points $P, Q \in \mathbb{R}^n$, the vector from P to Q is

$$\overrightarrow{PQ} = Q - P$$

Component-wise addition and scalar multiplication are:

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}, \quad \lambda \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \lambda a_1 \\ \vdots \\ \lambda a_n \end{pmatrix}.$$

1.2 Length and unit vectors

The Euclidean length of $\vec{a} = (a_1, \dots, a_n)$ is

$$\|\vec{a}\|_2 = \sqrt{a_1^2 + \dots + a_n^2}$$

If $\vec{a} \neq \vec{0}$, its normalized vector is

$$\hat{\vec{a}} = \frac{\vec{a}}{\|\vec{a}\|_2}.$$

The standard basis vectors of $\mathbb{R}^3$ are and $\|\vec{e}_1\| = \|\vec{e}_2\| = \|\vec{e}_3\| = 1$.

$$\vec{e}_1 = \vec{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \vec{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_3 = \vec{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

1.3 Linear combinations and span

A sum of scalar multiples of vectors is called a linear combination:

$$\lambda_1 \vec{v}_1 + \dots + \lambda_k \vec{v}_k = \sum_{i=1}^k \lambda_i \vec{v}_i$$

The span is the set of all such linear combinations:

$$\text{span}(\vec{v}_1, \dots, \vec{v}_k) = \left\{ \sum_{i=1}^k \lambda_i \vec{v}_i \mid \lambda_i \in \mathbb{R} \right\}$$

Every $\vec{x} \in \mathbb{R}^n$ can be expressed using the standard basis:

$$\vec{x} = (x_1, \dots, x_n) = x_1 \vec{e}_1 + \dots + x_n \vec{e}_n.$$

$$\vec{a} \cdot \vec{b} = \vec{a}^T \vec{b} = \sum_{i=1}^n a_i b_i = \|\vec{a}\| \|\vec{b}\| \cos \varphi$$

1.4 Dot product

For $\vec{a}, \vec{b} \in \mathbb{R}^n$:

The dot product satisfies

$$\begin{aligned} \vec{a} \cdot \vec{a} &\geq 0, & \vec{a} \cdot \vec{a} = 0 &\iff \vec{a} = \vec{0}, \\ \vec{a} \cdot \vec{b} &= \vec{b} \cdot \vec{a}, & (\vec{a} + \vec{b}) \cdot \vec{c} &= \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}, \\ (\lambda \vec{a}) \cdot \vec{b} &= \lambda(\vec{a} \cdot \vec{b}). \end{aligned}$$

Two vectors are orthogonal exactly when

$$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$$

For non-zero vectors, the angle between them is

$$\varphi = \arccos \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} \right).$$

If α, β, γ are the angles between a vector and the x -, y -, and z -axes, then

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.$$

Mechanical work by a constant force is

$$W = \vec{F} \cdot \vec{s} = \|\vec{F}\| \|\vec{s}\| \cos \varphi.$$

1.5 Cross product (vector product)

For $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$:

$$\vec{u} \times \vec{v} = (u_2 v_3 - u_3 v_2, -(u_1 v_3 - u_3 v_1), u_1 v_2 - u_2 v_1)$$

Equivalently, using reduced determinants:

$$\vec{u} \times \vec{v} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, -\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right)$$

The cross product is perpendicular to both vectors and

$$\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \varphi$$

is the area of the parallelogram they span. Furthermore,

$$\begin{aligned} \vec{u} \times \vec{v} &= -(\vec{v} \times \vec{u}), \\ \vec{u} \times (\vec{v} + \vec{w}) &= \vec{u} \times \vec{v} + \vec{u} \times \vec{w}, \\ (\lambda \vec{u}) \times \vec{v} &= \lambda(\vec{u} \times \vec{v}), \\ \vec{u} \times \vec{v} = \vec{0} &\iff \vec{u}, \vec{v} \text{ are collinear (or one is zero).} \end{aligned}$$

2 Matrices

2.1 Definitions and special matrices

An $m \times n$ matrix has m rows and n columns:

$$A = (a_{ij}) \in \mathbb{R}^{m \times n}.$$

A matrix is square if $m = n$. A matrix in $\mathbb{R}^{n \times 1}$ is a column vector, and one in $\mathbb{R}^{1 \times n}$ is a row vector.

Important special matrices are:

$$O = (0)_{m \times n}, \quad I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix},$$

upper and lower triangular matrices, and diagonal matrices $D = \text{diag}(d_1, \dots, d_n)$.

The transpose interchanges rows and columns:

$$(A^T)_{ij} = a_{ji}.$$

A square matrix is symmetric if

$$A^T = A$$

2.2 Addition and scalar multiplication

For matrices of equal size:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

The usual associative, commutative, distributive, zero, and additive inverse laws apply.

2.3 Matrix multiplication

If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, then $AB \in \mathbb{R}^{m \times p}$ with

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

The multiplication rules are

$$\begin{aligned} (AB)C &= A(BC), & (A+B)C &= AC + BC, \\ A(B+C) &= AB + AC, & \lambda(AB) &= (\lambda A)B = A(\lambda B), \\ AI &= IA = A, & (AB)^T &= B^T A^T. \end{aligned}$$

In general, $AB \neq BA$.

3 Linear Systems and Gaussian Elimination

A linear system can be written as $A\vec{x} = \vec{b}$ where A is the coefficient matrix, $\vec{x}$ is the vector of unknowns, and $\vec{b}$ is

$$A\vec{x} = \vec{b}$$

the right-hand side. Its augmented matrix is $[A \mid \vec{b}]$.

3.1 Elementary row operations

Gaussian elimination uses:

1. interchange two rows;
2. multiply a row by a non-zero scalar;
3. add a multiple of one row to another row.

The objective is row echelon form, with each pivot farther right than the pivot above it and zero entries below every pivot:

$$\left[\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array} \right].$$

Gauss-Jordan elimination additionally produces zeros above each pivot, giving reduced row echelon form.

3.2 Solution cases

- A pivot in every variable column and no contradiction: one unique solution.
- At least one free variable and no contradiction: infinitely many solutions.
- A row $[0 \ \cdots \ 0 \ | \ c]$ with $c \neq 0$: no solution.

3.3 Homogeneous and inhomogeneous systems

The homogeneous system

$$A\vec{x} = \vec{0}$$

always has the trivial solution $\vec{x} = \vec{0}$. If free variables exist, it also has infinitely many non-trivial solutions.

For a consistent inhomogeneous system $A\vec{x} = \vec{b}$, every solution has the form

$$\vec{x} = \vec{x}_p + \vec{x}_h, \quad A\vec{x}_p = \vec{b}, \quad A\vec{x}_h = \vec{0}$$

4 Determinants and Inverse Matrices

4.1 Determinant

For a 2×2 matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

For $A \in \mathbb{R}^{3 \times 3}$, Sarrus' rule gives:

$$\left[\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array} \right]$$

$$\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

For a triangular matrix,

$$\det A = \prod_{i=1}^n a_{ii}.$$

For a general square matrix, reduce it to triangular form while tracking:

- swapping two rows changes the sign of the determinant;
- multiplying a row by λ multiplies the determinant by λ ;
- adding a multiple of one row to another does not change the determinant.

Important rules are

$$\det(A^T) = \det A,$$

$$\det(\lambda A) = \lambda^n \det A,$$

$$\det(AB) = \det A \det B,$$

$$\det(A^{-1}) = \frac{1}{\det A}.$$

A square matrix is singular exactly when $\det A = 0$.

4.2 Inverse matrix

A square matrix A is invertible if there is a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n$$

For a 2×2 matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0$$

In general,

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A), \quad \det A \neq 0.$$

It can also be computed by Gauss-Jordan elimination:

$$[A \mid I_n] \sim [I_n \mid A^{-1}].$$

If A is invertible, the unique solution of $A\vec{x} = \vec{b}$ is

$$\vec{x} = A^{-1}\vec{b}$$

5 Vector Spaces, Subspaces, and Bases

5.1 Subspaces

A subset $U \subseteq \mathbb{R}^n$ is a vector subspace (or linear subspace) if:

$$U1) \vec{0} \in U;$$

$$U2) \vec{u}, \vec{v} \in U \implies \vec{u} + \vec{v} \in U;$$

$$U3) \vec{u} \in U, \lambda \in \mathbb{R} \implies \lambda \vec{u} \in U.$$

The span of any set of vectors is a subspace. In particular, $\operatorname{span}(\vec{v}_1, \dots, \vec{v}_k)$ is the smallest subspace containing all of those vectors.

5.2 Linear independence

Vectors $\vec{v}_1, \dots, \vec{v}_k$ are linearly independent if Otherwise they are linearly dependent.

$$\lambda_1 \vec{v}_1 + \dots + \lambda_k \vec{v}_k = \vec{0} \implies \lambda_1 = \dots = \lambda_k = 0$$

Two non-zero vectors are dependent exactly when one is a scalar multiple of the other. A single vector is independent exactly when it is non-zero.

If the vectors are columns of a matrix A , they are independent exactly when the homogeneous system $A\vec{x} = \vec{0}$ has only the trivial solution.

5.3 Basis and dimension

The vectors $\vec{v}_1, \dots, \vec{v}_k \in U$ form a basis of U if they:

1. span U ;
2. are linearly independent.

Every vector in U then has a unique representation as a linear combination of the basis vectors.

The dimension $\dim U$ is the number of vectors in any basis of U . In particular,

$$\dim \mathbb{R}^n = n, \quad \dim\{\vec{0}\} = 0.$$

If $\dim U = d$ and $\vec{v}_1, \dots, \vec{v}_d \in U$, then the following are equivalent:

$$\{\vec{v}_1, \dots, \vec{v}_d\} \text{ is a basis of } U \iff \begin{cases} \vec{v}_1, \dots, \vec{v}_d \text{ are independent,} \\ U = \text{span}(\vec{v}_1, \dots, \vec{v}_d). \end{cases}$$

With exactly d vectors, it is enough to verify either independence or spanning.

5.4 Affine subspaces

An affine subspace is a translated vector subspace: where $\vec{v} \in \mathbb{R}^n$ and $U \subseteq \mathbb{R}^n$ is a subspace. Its dimension

$$A = \vec{v} + U = \{\vec{v} + \vec{u} \mid \vec{u} \in U\}$$

is $\dim A = \dim U$.

$\dim A$	geometric object
0	point
1	line
2	plane
$n - 1$	hyperplane

A consistent solution set of $A\vec{x} = \vec{b}$ is an affine subspace: $\vec{x}_p + \ker A$.

6 Linear and Affine Maps

6.1 Linear maps

A map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if, for all $\vec{u}, \vec{v} \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$, Equivalently, it is additive and homogeneous:

$$L(\lambda\vec{u} + \mu\vec{v}) = \lambda L(\vec{u}) + \mu L(\vec{v})$$

$$L(\vec{u} + \vec{v}) = L(\vec{u}) + L(\vec{v}), \quad L(\lambda\vec{u}) = \lambda L(\vec{u}).$$

Every linear map satisfies $L(\vec{0}) = \vec{0}$.

Every linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has a matrix $A \in \mathbb{R}^{m \times n}$ such that

$$L(\vec{x}) = A\vec{x}.$$

In the standard bases, its columns are

$$A = \begin{bmatrix} L(\vec{e}_1) & \cdots & L(\vec{e}_n) \end{bmatrix}$$

6.2 Common transformations

Counter-clockwise rotation through angle θ :

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Clockwise rotation through angle θ :

$$R_{-\theta} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

6.3 Affine maps

A map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is affine if for a linear map L , a matrix A , and a translation vector $\vec{v}$. It is linear exactly

$$F(\vec{x}) = L(\vec{x}) + \vec{v} = A\vec{x} + \vec{v}$$

when $\vec{v} = \vec{0}$.

6.4 Inverse maps

A linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible if there is a map L^{-1} such that

$$L^{-1}(L(\vec{x})) = \vec{x}, \quad L(L^{-1}(\vec{y})) = \vec{y}.$$

If $L(\vec{x}) = A\vec{x}$, then L is invertible exactly when $\det A \neq 0$.

$$L^{-1}(\vec{x}) = A^{-1}\vec{x}$$

7 Kernel, Image, Rank, and Nullity

For $L(\vec{x}) = A\vec{x}$:

$$\ker L = \{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = \vec{0}\}, \quad \text{im } L = \{A\vec{x} \mid \vec{x} \in \mathbb{R}^n\}$$

If $A = [\vec{a}_1 \ \dots \ \vec{a}_n]$, then

$$\text{im } A = \text{span}(\vec{a}_1, \dots, \vec{a}_n).$$

To obtain a basis of the image, row-reduce A and select the original columns corresponding to pivot columns.

The rank and nullity are

$$\text{rank } A = \dim(\text{im } A), \quad \text{nullity } A = \dim(\ker A).$$

The rank is the number of pivots; the nullity is the number of free variables. For $A \in \mathbb{R}^{m \times n}$:

$$\text{rank } A + \text{nullity } A = n$$

7.1 Consequences for systems

For $A \in \mathbb{R}^{m \times n}$:

$$\ker A = \{\vec{0}\} \iff \text{nullity } A = 0 \iff \text{rank } A = n \iff \text{the columns of } A \text{ are independent.}$$

Furthermore, $\text{im } A = \mathbb{R}^m \iff \text{rank } A = m \iff A\vec{x} = \vec{b}$ is solvable for every $\vec{b} \in \mathbb{R}^m$.

7.2 Equivalent conditions for square matrices

For $A \in \mathbb{R}^{n \times n}$, the following are equivalent:

1. A is invertible;
2. $\det A \neq 0$;
3. $\ker A = \{\vec{0}\}$;
4. $\text{rank } A = n$;
5. $\text{im } A = \mathbb{R}^n$;
6. the rows and columns of A each form a basis of $\mathbb{R}^n$;
7. $A\vec{x} = \vec{0}$ has only the trivial solution;
8. $A\vec{x} = \vec{b}$ has the unique solution $\vec{x} = A^{-1}\vec{b}$, $\forall \vec{b} \in \mathbb{R}^n$.

8 Eigenvalues and Eigenvectors

Let $A \in \mathbb{R}^{n \times n}$. A scalar λ is an eigenvalue of A if there is a non-zero vector $\vec{v}$ such that $A\vec{v} = \lambda\vec{v}$, $\vec{v} \neq \vec{0}$.

The vector $\vec{v}$ is an eigenvector corresponding to λ . Every non-zero scalar multiple of an eigenvector is another eigenvector for the same eigenvalue.

8.1 Eigenspaces

The eigenspace associated with λ is

$$E_\lambda = \ker(A - \lambda I) = \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \lambda\vec{v}\}$$

It contains the zero vector; its non-zero elements are the eigenvectors associated with λ .

8.2 Computing eigenvalues and eigenvectors

1. Form the characteristic polynomial

$$p_A(\lambda) = \det(A - \lambda I).$$

2. Solve $p_A(\lambda) = 0$. Its roots are the eigenvalues.

3. For each eigenvalue λ_i , solve

$$(A - \lambda_i I)\vec{v} = \vec{0}$$

to find a basis of E_{λ_i} .

9 Diagonalization

A square matrix A is diagonalizable if there is a basis of $\mathbb{R}^n$ consisting of eigenvectors of A . If $\vec{v}_1, \dots, \vec{v}_n$ are independent eigenvectors with eigenvalues $\lambda_1, \dots, \lambda_n$, define

$$S = [\vec{v}_1 \quad \dots \quad \vec{v}_n], \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Then

$$S^{-1}AS = D \iff A = SDS^{-1}$$

An $n \times n$ matrix is diagonalizable exactly when the sum of the dimensions of its eigenspaces is n . In particular, n distinct eigenvalues guarantee diagonalizability.

Diagonalization simplifies powers:

$$A^k = SD^kS^{-1}.$$

10 Norms, Metrics, and Orthogonality

10.1 Norms

A norm satisfies, for all vectors $\vec{x}, \vec{y}$ and scalars λ :

$$\begin{aligned} \|\vec{x}\| &\geq 0, \\ \|\vec{x}\| &= 0 \iff \vec{x} = \vec{0}, \\ \|\lambda\vec{x}\| &= |\lambda|\|\vec{x}\|, \\ \|\vec{x} + \vec{y}\| &\leq \|\vec{x}\| + \|\vec{y}\|. \end{aligned}$$

Common norms are

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}, \quad \|\vec{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

10.2 Metrics

A metric $d : M \times M \rightarrow \mathbb{R}$ satisfies:

$$\begin{aligned}d(x, y) &\geq 0, \\d(x, y) &= 0 \iff x = y, \\d(x, y) &= d(y, x), \\d(x, y) &\leq d(x, z) + d(z, y).\end{aligned}$$

A norm induces the metric $d(\vec{x}, \vec{y}) = \|\vec{x} - \vec{y}\|$.

10.3 Orthogonal and orthonormal bases

A basis $\vec{b}_1, \dots, \vec{b}_k$ is orthogonal if

$$\vec{b}_i \cdot \vec{b}_j = 0 \quad (i \neq j).$$

It is orthonormal if additionally

$$\|\vec{b}_i\| = 1.$$

For an orthonormal basis, the coordinates of $\vec{v}$ are particularly simple:

$$\vec{v} = \sum_{i=1}^k (\vec{v} \cdot \vec{b}_i) \vec{b}_i.$$

10.3.1 Example

Show that b_1, b_2 is an ONB of the plane $E = \text{span}(b_1, b_2)$.

$$\begin{aligned}\|b_1\| &= 1 \quad ; \quad \|b_2\| = 1 \\b_1 \cdot b_2 &= 0 \\b_1 &\neq \alpha b_2, \alpha \in \mathbb{R} \quad \text{set the system of equation to find } \alpha\end{aligned}$$

11 Orthogonal Projections and Gram-Schmidt

11.1 Projection onto a line

The orthogonal projection of $\vec{v}$ onto $L = \text{span}(\vec{b})$, $\vec{b} \neq \vec{0}$, is If $\|\vec{b}\| = 1$, then

$$\text{proj}_{\vec{b}}(\vec{v}) = \frac{\vec{v} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \vec{b}$$

$$\text{proj}_{\vec{b}}(\vec{v}) = (\vec{v} \cdot \vec{b}) \vec{b}.$$

The error vector is orthogonal to the line:

$$\vec{v} - \text{proj}_{\vec{b}}(\vec{v}) \perp \vec{b}.$$

11.2 Projection onto a subspace

If $\vec{b}_1, \dots, \vec{b}_k$ is an orthonormal basis of U , then The residual $\vec{v} - \text{proj}_U(\vec{v})$ lies in $U^\perp$.

$$\text{proj}_U(\vec{v}) = \sum_{i=1}^k (\vec{v} \cdot \vec{b}_i) \vec{b}_i$$

11.3 Orthogonal decomposition

Given a plane $E = \text{span}(b_1, b_2)$, to find the orthogonal decomposition of a vector v w.r.t E , where $p, w \in \mathbb{R}^3$:

$$v = p + w, p \in E, w \perp E$$

$$\begin{cases} v = \alpha b_1 + \beta b_2 + w \\ w \perp E \end{cases} \Rightarrow \begin{cases} v = \alpha b_1 + \beta b_2 + w \\ w \cdot b_1 = 0 \\ w \cdot b_2 = 0 \end{cases} \Rightarrow \begin{cases} v = \alpha b_1 + \beta b_2 + w \\ w = k(b_1 \times b_2) \end{cases}$$

Thus,

$$v = \alpha b_1 + \beta b_2 + k(b_1 \times b_2)$$

If b_1, b_2 are ONB, $\alpha, \beta, k = 1$.

11.4 Gram-Schmidt process

Starting from a basis $\vec{v}_1, \dots, \vec{v}_k$, construct an orthogonal basis:

$$\begin{aligned} \vec{u}_1 &= \vec{v}_1, \\ \vec{u}_2 &= \vec{v}_2 - \text{proj}_{\vec{u}_1}(\vec{v}_2), \\ \vec{u}_j &= \vec{v}_j - \sum_{i=1}^{j-1} \text{proj}_{\vec{u}_i}(\vec{v}_j). \end{aligned}$$

Normalize the resulting vectors: Then $\vec{b}_1, \dots, \vec{b}_k$ is an orthonormal basis of the same subspace.

$$\vec{b}_i = \frac{\vec{u}_i}{\|\vec{u}_i\|}, \quad i = 1, \dots, k$$

11.5 Algorithm in a nutshell

$$\text{Step 1. } w_1 \rightarrow v_1 = \frac{w_1}{\|w_1\|}$$

$$\text{Step 2. } v'_2 = w_2 - \underbrace{(w_1 \cdot v_1) v_1}_{P_{\text{span}(v_1)}(w_2)}$$

$$\text{Step 3. } v_2 = \frac{v'_2}{\|v'_2\|}$$

$$\text{Step 4. } v'_3 = w_3 - \underbrace{(w_3 \cdot v_1) v_1 - (w_3 \cdot v_2) v_2}_{P_{\text{span}(v_1, v_2)}(w_3)}$$

$$\text{Step 5. } v_3 = \frac{v'_3}{\|v'_3\|}$$

12 Spectral Theorem

Every real symmetric matrix $A = A^T$:

- has only real eigenvalues;
- has mutually orthogonal eigenspaces for distinct eigenvalues;
- has an orthonormal basis of eigenvectors;
- is orthogonally diagonalizable.

If the orthonormal eigenvectors are the columns of V and their eigenvalues form D , then $V^{-1} = V^T$ and

$$V^T A V = D \quad \Longleftrightarrow \quad A = V D V^T$$